

C5

Computer interface

List of contents

1. General notes	5
2. Interfaces	6
2.1 RS232 connection.....	6
2.2 Network connection through 10BaseT Ethernet	7
3. Transmission protocol	8
3.1. Network protocol (NTP).....	10
3.1.1. Login telegram	11
3.1.2. Data telegram	11
4. The C5 main computer protocol.....	12
4.1. Parameters	12
4.1.1. Parameter name conventions	12
4.1.2. Parameter handle	13
4.2. Telegram mode.....	14
4.3. Commands.....	15
4.3.1. Command I (Init): Interface reset.....	15
4.3.2. Command V (Version): Read C5 version number	16
4.3.3. Command Q (Query Handle): Request parameter handle	17
4.3.4. Command R (Read): Parameter values reading	18
4.3.5. Command W (Write): Parameter writing	19
4.3.6. Command S (Set Request): Set-request for parameters	20
4.3.7. Command C (Clear Request): Cancel request for parameter	21
4.3.8. Command N (New Values): Read altered parameters	22
4.3.9. Command A (Alarm): Read and reset alarms	23
4.3.10. Command M (Machine info): Read machine info	25
4.3.11. The parameters of the C5	26
4.4. Parameter values.....	26
4.5. Parameter elements.....	26
4.6. C4 compatibility.....	28
4.7. Parameter descriptions	29
4.7.1. Basic parameters	30
4.7.2. Temperatures.....	31
4.7.3. Items of status information.....	33
4.7.4. Operating hours counter	34
4.7.5. Seven-day timer.....	35
4.7.6. Gravimetric system (option)	36
4.7.7. Saw-cut control and pipe-wall thickening (option)	37
4.7.8. Ultrasonic wall-thickness gauging (option)	38
4.7.9. Downstream for sheet lines (option)	39
4.7.10. Melt pump (option)	40
5. Programming a host interface.....	41
5.1. General notes	41
5.2. Connection setup	42
5.3. Data exchange.....	44
5.4. Severed connection	45

5.5. Programming a TCP/IP interface	46
5.5.1. Example program in C	46
5.5.2. .NET example program in C#.....	49
6. Establishing the network connection	51
6.1. Overview	51
6.2. Wiring.....	52
Assignment and cable colour codes for RJ45 plugs	52
Assignment of RJ45 plugs for Ethernet (network cable)	52
6.3. Connection to the C5	53
6.3.1. Cable connection	53
6.3.2. Network configuration	54
6.4. Network hub / switch	55
6.5. Connection to remote computer.....	56
6.5.1. Network installation	56
6.6. Testing the network connection	56

1. General notes

The C5 computer interface has been provided for reading actual values, switching points, items of status information, alarms and setting data for the C5 control system. Setting data can also be entered there.

It provides an opportunity for optimising the setting data of several machines at a central point, where these data can also be administered; actual values and alarms can be monitored from here as well.

This documentation uses various abbreviations, defined as follows:

C5	C5 extruder control system
ZR	Higher-level main computer
Host	Higher-level main computer
Network	10/100BaseT Ethernet
NTP	Network Transport Protocol

Reprinting, the production of copies and translations, in whole or in part, is permitted only with our prior consent in writing, and by quoting the source.

We reserve the rights for technical alterations.

2. Interfaces

2.1 RS232 connection

For the RS232 connection, both components are directly connected to each other via a serial cable.

This type of connection is no longer supported for the C5.

2.2 Network connection through 10BaseT Ethernet

The C5 supports the master computer interface via an **Ethernet** network of type **10/100BaseT**. This is a 10/100 MBit Ethernet with Twisted Pair cables (UTP or STP) and with RJ45 connecting plugs (Western plugs) at both ends.

RJ45 pin assignment:

Pin:	Abbreviation	Meaning
1	TD+	Transmit data +
2	TD-	Transmit data -
3	RD+	Receive data +
6	RD-	Receive data -

The network has a **star-shaped** structure, and each C5 machine must have a connection to the **hub** (star distributor). The hub forms the central exchange.

The **length of the cables** from and to the hub must not exceed **100 meters**. To help prevent noise and interference, STP cables (Shielded Twisted Pair) should be used in preference to UTP cables (Unshielded Twisted Pair).

The average response time for the network connection is approx. 200ms.
The maximum response time is 1 sec.

3. Transmission protocol

The **network transport protocol (NTP)** is used as the transmission protocol. This subprotocol is based on TCP/IP and ensures secure transmission.

The C5 control system is always passive. It responds only to a command received from the master computer (master-slave principle).

The data contents are displayed in this document according to the following syntax:

ASCII characters:

These are set in inverted commas, or when several characters have been strung together to form a string, they are set in quotation marks.

e.g.: 'C' or "TEXT"

The binary value of an ASCII character may also be shown in hexadecimal form. Thus, for instance, the expression 'C' corresponds to the hex value 0x43.

In telegrams, only ASCII characters that can be displayed are transmitted (0x20 ... 0xFF)

All control codes (smaller than 0x20) are used for controlling the data transfer. They must not be used within a telegram. In the subsequent text, control codes are indicated by a preceding caret character (e.g. ^C for 0x03).

ASCII control characters:

Abbreviation of the ASCII character in capital letters. The following special ASCII characters are used:

STX	Start of Transmission (0x02, ^B)
SOH	Start of Header (0x01, ^A)
ETX	End of Transmission (0x03, ^C)
BS	Backspace (0x08, ^H)
LF	Line Feed (0x0A, ^J)
CR	Carriage Return (0x0D, ^M)
DC2	Device Control 2 (0x12, ^R)
CAN	Cancel (0x18, ^X)
DEL	Delete (0x7F)
NUL	Null character (0x00)

The transmission protocols are described on the following pages. The individual C5 master computer commands are specified in the chapter on "Commands". The commands that are possible are identical for all transmission protocols.



All commas (,) in the command strings on the following pages are intended only to separate the different components of the command string and are **not** part of the command string.

3.1. Network protocol (NTP)

The network protocol NTP is used when the C5 machines are connected via Ethernet.

The **TCP/IP protocol** serves as the basic protocol for the network connection. The C5 control system expects a TCP/IP network environment in a class C network.

The **network number** (IP address) is set in the visualisation program on the C5 machine. The network numbers must be assigned in accordance with TCP/IP standards.

The packets are transmitted as streams (TCP protocol) at **port 18901**. Since this method already ensures error-free transmission, no additional checksum calculation is required. The NTP protocol expands the data to be transmitted only by the addition of information concerning the length and type of data.

The NTP frame has the following structure:

Bit 7		Bit 0	
0	Ctrl	Data type	
Data length (MSB)			
Data length (LSB)			
Data (with length 0..65535)			

Details of fields:

Data type byte:

Bit 7 of the data type byte must always be 0. This bit is reserved for future expansion.

The **Ctrl** bit (Bit 6) is only set if it is a control telegram for the transmission layer. In the case of the C5 computer interface, 0 should always be specified here.

The **Data type** describes the type of telegram, in other words which kind of data have been transmitted. There are two types: 0 is used for data telegrams and 1 is used for login telegrams.

Data length:

Two bytes which describe the length of the subsequent data in the telegram. First transmit the most significant byte (MSB) and then the least significant byte (LSB). Theoretically up to 65535 bytes can be transmitted in one telegram. At present, telegrams of **max. 8 Kb** are transmitted.

Data:

Within the data block, any characters and data may be used.

3.1.1. Login telegram

After a network connection is established between the C5 and the host, the host must log in to the C5. This is done using a LOGIN telegram.

The login telegram is of data type 1 and provides a user name and a password, each terminated with a NULL character. At the moment, the user name and password are not evaluated by the system, and can be omitted. It is sufficient to send the following three bytes as the login telegram.

01H (login)
00H data length (MSB)
00H data length (LSB)

3.1.2. Data telegram

Data telegrams cannot be exchanged until you are logged on. For data telegrams, data type 0 is always required.

Thus a data telegram always starts with a 00H followed by two bytes specifying the data length, which are then followed by the real data.

00H (data telegram)
Data length (MSB)
Data length (LSB)
Data (max. 8k bytes)



A list of the commands and their functions is provided in Chapter 5.3.

4.The C5 main computer protocol

4.1. Parameters

4.1.1. Parameter name conventions

Parameter names are used to identify parameters when reading or writing values. These parameter names are firmly defined for all C5 extrusion lines.

Parameter names always consist of three name parts, which are separated from each other by a full stop:

<parameter object>.<parameter element>.<parameter value>

All functionally related parameters are united in a <parameter object>. Thus parameter object Ext1_BarrelZone01, for instance, contains all parameters that are connected with the barrel heating zone 1 of extruder 1.

A single element of the parameter object is designated using the <parameter element>. The number of possible parameter elements within an object is subject to its type. In the case of a barrel heating zone, the following parameter elements are available among others:

Set	Set value
Act	Actual value
EDSet	ED-position (set value programming)
EDAct	Current heating value/cooling value
Xph	Proportional value for heating
Tah	Heating-controller scanning frequency
Tnh	Integral action time for heating
...	

The <parameter value> specified defines which information of the parameter element is to be read or written. Conventional parameter values are:

Val	Current value
Min	Minimum value
Max	Maximum value
LTxt	Parameter long text
Dim	Dimension (= physical unit)
...	

To read out the maximum value of the actual value of barrel zone 1, for example, the complete parameter name must read "Ext1_BarrelZone01.Act.Max".

In the C5, parameter names are not case-sensitive.

4.1.2. Parameter handle

A parameter name can be up to 64 characters long. In order to reduce the quantity of transmission data, the C5 computer interface has the ability to allocate an identification number (parameter handle) to each parameter name. These parameter handles consist of **4-figure hexadecimal numbers**, prefixed by the '\$' character (0x24). This means that the parameter name can be reduced from 64 to just 5 characters.

Example: "\$0001" or "\$001B"

The assignment of parameter handle to parameter name is always performed by the C5. The Q command must be used to query a parameter handle.

Once the C5 has been asked for a parameter handle, all access to this parameter can be controlled through the handle from then on.

All queried parameter handles are consecutively numbered from 0 onwards so that the parameter handle can be used directly as an index of an administration list.

4.2. Telegram mode

There are two different telegram modes for the C5 main computer interface: the standard mode and the extended mode.

In **standard mode** up to 4 parameters can be transmitted within one telegram.

In **extended mode** up to 16 parameters can be transmitted within one telegram. In the extended mode it is important to ensure that the interface reception buffer is large enough to accommodate the larger number of parameters. This requires a buffer capacity of 8 Kb.

The extended mode is especially recommended for network connections, since the reduced number of telegrams results in a higher throughput rate than shortening the length of telegrams.

The telegram mode can be selected using the INIT command.

4.3. Commands

For explaining the commands, only the data-section of the program will be displayed in the following. Depending on the transmission protocol, the data may also have to be provided with the corresponding telegram frame (see Chapter 3).

4.3.1. Command I (Init): Interface reset

Function: All parameter requests, which may possibly have been set previously, are deleted.
All allocated parameter handles are erased and are no longer valid.
This command should be carried out once before the start of every interface communication.

A distinction is made between "standard" and "extended" operating modes during initialisation.

In "standard mode" up to 4 parameters per telegram can be transmitted.

In "extended mode" up to 16 parameters per telegram can be sent. In "extended mode" the reception buffer must have a maximum capacity 8 Kb.

ZR → C5: 1) "I:" Initialise and activate "standard mode"
2) "I:X" Initialise and activate "extended mode"

C5 → ZR: "I:.*"

Faults: None.

Example: ZR → C5: "I:"
C5 → ZR: "I:.*"

4.3.2.Command V (Version): Read C5 version number

Function: With this command, the version number of the C5 visualisation program can be read. The version number is always shown in the format of xx.xx.xx-xxxx.

ZR → C5: "V:"

C5 → ZR: "V:xx.xx.xx-xxxx"

Faults: None.

Example: ZR → C5: "V:"
C5 → ZR: "V:01.09.00-0000"



The version number format is different from the C4 format!

4.3.3.Command Q (Query Handle): Request parameter handle

Function: With this command, identification numbers (so-called parameter handles) can be allocated to each C5 parameter name. These handles are displayed as 4-figure hex numbers, with a '\$'-sign as prefix.

ZR → C5: "Q:p1| p2 | ... |pn"

p1,p2,...,p4 Parameter name
Up to 4 parameter handles (up to 16 in extended mode) can be requested simultaneously.

C5 → ZR: "Q:h1| h2 | ... |hn"

h1,h2,...,h4 Parameter handles (see Chapter 3.3.2)
The parameter handles are shown as 4-figure hexadecimal in the shape of an ASCII-string with a '\$'-sign prefix.
Should the parameter name prove to be wrong, or the parameter is non-existent, a question mark is returned instead of the parameter handle.

Example: ZR → C5: "Q:Ext1_ScrewSpeed.Set.Val|XYZ"
 C5 → ZR: "Q:\$0000|?"

Handle \$0000 is returned for the first parameter. The second one is invalid, however and is therefore acknowledged with a question mark.

4.3.4.Command R (Read): Parameter values reading

Function: With this command set values, actual values, limiting values, as well as texts of parameters can be read-out.

ZR → C5: "Q:p1| p2 | ... |pn"

p1,p2,...,pn Parameter names or parameter handles
Up to 4 parameter values (up to 16 in extended mode) can be read simultaneously.

C5 → ZR: "R:v1| v2 | ... |vn"

v1,v2,...,vn Parameter values
Parameter values are displayed as decimal figures in the shape of an ASCII-string.
Should the parameter name/the parameter handle be wrong, or the parameter be non-existent, a question mark is returned instead of the parameter value.

Example: ZR → C5: "R:\$0000|Ext1_FeedSpeed01.Set.Val|XYZ"
 C5 → ZR: "R:16,1|88,7|?"

The value is returned for the first two parameters, the third one is invalid, however and is therefore acknowledged with a question mark. A handle had already been requested for the first parameter and has here been used instead of the name.

4.3.5.Command W (Write): Parameter writing

Function: Parameter set values can be written to the C5 with this command.

ZR → C5: "W:p1=v1| p2=v2 | ... |pn=vn"

p1,p2,...,pn	Parameter names or parameter handles Up to 4 parameter (up to 16 in extended mode) values can be written simultaneously.
v1,v2,...,vn	Parameter values: the parameter values are displayed as decimal figures in the shape of an ASCII-string.

C5 → ZR: "W:q1| q2 | ... |qn"

q1,q2,...,qn	Acknowledgement. The following characters can be returned as acknowledgement: * (0x2A): The parameter value has been written successfully. + (0x2B): The parameter value is too large.- (0x2D): The parameter value is too small ? (0x3F): The parameter name/the parameter handle is wrong or the parameter is non-existent or the parameter cannot be written.
--------------	---

Example: ZR → C5:
"W:Ext1_ScrewSpeed.Set.Val=20,1|Ext1_Feed_Speed01.Set.Val=500"
C5 → ZR: "W:*|+"

The first parameter has been written successfully, the second parameter value is too large.



Parameters that can be written become effective immediately in the C5 so that they can influence the extrusion process at once. Therefore, any access by writing via the computer interface must be treated with the greatest care and should be avoided, when production is running.

4.3.6.Command S (Set Request): Set-request for parameters

Function: This command allows frequently read parameters to be included in a request list. By selecting the N-command, it is then very easy to read-out all parameters, that are contained within this request list and which have changed since the last reading procedure. This function dispenses with the elaborate polling of a multitude of parameters.

ZR → C5: "S:p1| p2 | ... |pn"

p1,p2,...,pn Parameter names or parameter handles
Up to 4 parameters (up to 16 in extended mode) can be requested simultaneously.

C5 → ZR: "S:q1| q2 | ... |qn"

q1,q2,...,qn Acknowledgement. The following characters can be returned as acknowledgement:
* (0x2A): The parameter value has been requested successfully.
? (0x3F): The parameter name/parameter handle is wrong or.
 The parameter is non-existent.

Example: ZR → C5: "S:\$0000|Ext1_Feed_Speed01.Set.Val|XYZ"
 C5 → ZR: "S:*|*|?"

The first two parameters have been requested successfully, the third parameter is invalid, however, and is therefore acknowledged with a question mark only. A handle had already been requested for the first parameter and has here been used instead of the name.

4.3.7.Command C (Clear Request): Cancel request for parameter

Function: With this command, requested parameters can be cancelled again from the request list. These parameters are then no longer taken into consideration with the N-command.

ZR → C5: "C:p1| p2 | ... |pn"

p1,p2,...,pn Parameter names or parameter handles
Up to 4 parameter requests (up to 16 in extended mode) can be cancelled simultaneously.

C5 → ZR: "C:q1| q2 | ... |qn"

q1,q2,...,qn Acknowledgement. The following characters can be returned as acknowledgement:
* (0x2A): The parameter value has been enabled successfully.
? (0x3F): The parameter name/parameter handle is wrong or.
 The parameter is non-existent.

Example: ZR → C5: "C:Ext1_Feed1_Speed.Set.VAL "
 C5 → ZR: "C:*

The parameters have been cancelled successfully from the request list.

4.3.8.Command N (New Values): Read altered parameters

Function: All parameters requested with the 'S' command are checked for any alteration in their value. Parameters, whose value has changed, are returned automatically, accompanied by the new current parameter value. A maximum of 4 parameters can be returned simultaneously in a telegram. By cyclic repeat performance of the N-command it is thus possible, to update changes in values continuously, without elaborate polling of each individual parameter.

ZR → C5: "N:"

C5 → ZR: "N:p1=v1| p2=v2 | ... |pn=vn"

p1,p2,...,pn Parameter names or parameter handles
 If the parameter's handle is known, then that handle is used every time, rather than the parameter name.
 Up to 4 parameter values (up to 16 in extended mode) can be returned simultaneously.

v1,v2,...,vn Parameter values: the parameter values are displayed as decimal figures in the shape of an ASCII-string.

If no parameter or value is returned, then the values for the requested parameters have not been altered.

Examples: ZR → C5: "N:"
 C5 → ZR: "N:\$0000=15,7|Ext1_Feed1_Speed.Act.VAL=100,1"

There are two parameter alterations. Parameter name/handle and parameter value will be returned in each case.

ZR → C5: "N:"
 C5 → ZR: "N:"

There are no parameter changes.

Note: If the reply telegram is faulty, it is possible to request a repeat of the last telegram with the control code DC2 (0x12). (Repeating the N-command would not produce the same reply-string).

4.3.9.Command A (Alarm): Read and reset alarms

Function: With the 'A' command you have four ways to operate with alarm information.

ZR → C5:

1)	"A:"	read all active alarms
2)	"A:COUNT"	read current number of active alarms
3)	"A:RESet"	reset alarms
4)	"A:PRIO"	read alarms and their priority
5)	"A:<alarm number>"	read corresponding alarm text

C5 → ZR:

1)	"A:a1 a2 .. an"
2)	"A:c"
3)	"A:*"
4)	"A:a1,p1 a2,p2 ... an,pn"
5)	"A:<alarm text>"

a1,a2,...an Alarm numbers for all alarms currently active
Theoretically, all existing alarm numbers can be returned. The number of alarms can be determined using the "A:COUNT" command.

p1,p2,...,pn Alarm priorities are no longer supported in the C5.

c Number of alarms:

<alarm text> Alarm text in the currently set language



The alarm numbers are different from those of the C4 control system. Alarm priorities are no longer supported in the C5.

Examples: ZR → C5: "A:"
C5 → ZR: "A:01000|15000|01501"

Currently, alarms 1000, 15000 and 1501 are active.

ZR → C5: "A:"
C5 → ZR: "A:"

No alarm active.

ZR → C5: "A:COUNT"
C5 → ZR: "A:3"

Currently 3 alarms active.

ZR → C5: "A:RESet"
C5 → ZR: "A:*"

alarms reset.

4.3.10.Command M (Machine info): Read machine info

Function: With the 'M' command you can read the machine name, the order number and the machine number.

ZR → C5: "M:"

C5 → ZR: "M:<machine name><order no.><machine no.>

Machine name, order number and machine number are separated by the delimiter. The length of the numbers is unlimited.

Example: ZR → C5: "M:"
C5 → ZR: "M: KMD 114-26|38001234|51001234"
(| = blank)



The reply string of the machine info is different from that of the C4!

4.3.11. The parameters of the C5

The following chapter lists all parameters that can be read-out or written to over the C5 computer interface.

The naming convention for C5 parameter names is described in Chapter 3.3.1.

As described there, each parameter name is composed of the following three name sections:

<object name>.<element name>.<value name>

The number of possible elements within an object and the names of these elements are subject to the object type.

The possible values of a parameter element on the other hand are a function of the element type (also called parameter type in subsequent text).

4.4.Parameter values

The C5 differentiates between various parameter types. Subject to parameter type, different values of the parameter can be requested. These values may be available in different formats. However, all formats are ASCII-strings on principle, with differentiation being made between:

Decimal	Represent decimal numbers, with a potential decimal point being represented by a comma (0x2C) (for example, "23,99")
Text	Any text, whereby the vertical separator line (0x7C) is not permitted as a character (for example, "Extruder")
Date	Specifies the date, with day, month and year divided by full stops '.' (e.g. "07.12.96")
Time	Specifies the time, with hours, minutes and seconds being divided by colons ':' (for example, "12:20:00"). The time can be displayed with or without seconds.
Duration	Specifies a duration in hours and minutes. Hours and minutes are separated by a colon (for example, "188:12")

4.5.Parameter elements

We differentiate between the following parameter element types:

1. Analog value parameter (ANALOG)

The following values are available with an analogue value parameter.

Name	Description	Format
Val	Current value of the parameters	Decimal
Min	Lower limiting value of the parameters	Decimal
Max	Upper limiting value of the parameter	Decimal
Dim	Physical unit in the language of the actual country	Text
STxt	Abbreviated parameter text in the actual language of the country	Text
LTxt	Parameter text in the current country's language	Text

2. Switching function / status (SWITCH)

The following values are available with a switching function:

Name	Description	Format
Val	Current value of the parameters	Decimal
Max	Maximum selection function value	Decimal
LTxt	Parameter text in the current country's language	Text

3. String parameter (STRING)

The following values are available with a string parameter:

Name	Description	Format
Val	Current string of the parameters	Text
Len	Maximum permissible length of the string	Decimal
LTxt	Parameter text in the current country's language	Text

4. Date/time parameter (DATE)

The following values are available with a date parameter:

Name	Description	Format
Val	Value as no. of seconds since 01.01.1980	Decimal
Date	Date in the format "dd.mm.yy"	Date
Time	Time in the "hh:mm" format	Time
LTxt	Parameter text in the current country's language	Text

5. Duration parameter (DURATION)

The following values are available with a time parameter:

Name	Description	Format
Val	Value as number of seconds	Decimal
Time	Duration as "hhh:mm" format	Duration
TimeSecs	Duration in seconds as "hhh:mm:ss" format	Duration
LTxt	Parameter text in the current country's language	Text



All parameter values with the exception of Val are "read only".

Whether the Val parameter value can also be written is subject to the parameter's use. For more information, refer to the parameter description in Chapter 5.3 in the "Mode" column of each parameter.

4.6. C4 compatibility

The C5 main computer interface protocol is approx. 95% similar to that of the C4 central computer interface.

Only very few commands cannot produce the exact same reply string of the C4 due to format changes in the C5.

However, we have ensured that changes to the interface are minimal.

The C4 and C5 interfaces can easily be distinguished by the version number format via the V command.

C4 version: "x.x.xx-xxx"

C5 version: "xx.xx.xx-xxxx"

In order to simplify parameter names for C4 connections already installed, a table of names has been created that translates the most important C4 parameters into C4 parameter names and thus enables operation of a C5 extruder via a C4 computer interface.

4.7.Parameter descriptions

This chapter describes all C5 parameters that can be accessed via the computer interface. Many of the parameters listed on the subsequent pages are linked to purchasing options, which means, that they are therefore only available, when the respective functions have been isolated.

Abbreviated prefixes to parameter texts, such as Ex1, Ex2, Ex3 denote the respective extruder, to which the parameter object has been allocated. The same applies to Dos1, Dos2, Dos3 (matched with the feed unit) and Abz1, Abz2 (associated with the haul-off).

The "**Type**" column indicates the parameter element types, which are described in Chapter 5.3. The type identifies the values, that can be requested from this parameter.

The "**Dim**" column indicates the physical unit, in which a parameter value can be read or written.

The "**Mode**" column indicates, if a parameter can only be read (R) or read and written (R / W).



Parameters that can be written become effective immediately in the C5 so that they can influence the extrusion process at once. Therefore, any access by writing via the computer interface must be treated with the greatest care and should be avoided, when production is running.

4.7.1. Basic parameters

Object	Element	Description	Type	Dim	Mode
Ext1_AutoVacuum	Set	Ex1: Screw torque for turning OFF the vacuum	Analog	[%]	R / W
Ext2_AutoVacuum	Set	Ex2: Screw torque for turning OFF the vacuum	Analog	[%]	R / W
Ext1_ScrewSpeed	Set	Set value Ex1: screw RPM	Analog	[rpm]	R / W
	Act	Actual value Ex1: screw RPM	Analog	[rpm]	R
Ext2_ScrewSpeed	Set	Set value Ex2: screw RPM	Analog	[rpm]	R / W
	Act	Actual value Ex2: screw RPM	Analog	[rpm]	R
Ext3_ScrewSpeed	Set	Set value Ex3: screw RPM	Analog	[rpm]	R / W
	Act	Actual value Ex3: screw RPM	Analog	[rpm]	R
Ext1_Feed1_Speed	Set	Extruder1: feed unit 1: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder1: feed unit 1: actual value feed unit RPM	Analog	[rpm]	R
Ext1_Feed2_Speed	Set	Extruder1: feed unit 2: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder1: feed unit 2: actual value feed unit RPM	Analog	[rpm]	R
Ext1_Feed3_Speed	Set	Extruder1: feed unit 3: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder1: feed unit 3: actual value feed unit RPM	Analog	[rpm]	R
Ext1_Feed4_Speed	Set	Extruder1: feed unit 4: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder1: feed unit 4: actual value feed unit RPM	Analog	[rpm]	R
Ext2_Feed1_Speed	Set	Extruder2: feed unit 1: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder2: feed unit 1: actual value feed unit RPM	Analog	[rpm]	R
Ext2_Feed2_Speed	Set	Extruder2: feed unit 2: set value feed unit RPM	Analog	[rpm]	R / W
	Act	Extruder2: feed unit 2: actual value feed unit RPM	Analog	[rpm]	R
Hoff1_Speed	Set	Set value Hoff1: haul-off speed	Analog	[m/min]	R / W
	Act	Actual value Hoff1: haul-off speed	Analog	[m/min]	R
Hoff2_Speed	Set	Set value Hoff2: haul-off speed	Analog	[m/min]	R / W
	Act	Actual value Hoff2: haul-off speed	Analog	[m/min]	R
Ext1_ScrewTorque	Act	Extruder1: Actual value screw torque	Analog	[%]	R
Ext2_ScrewTorque	Act	Extruder2: Actual value screw torque	Analog	[%]	R
Ext3_ScrewTorque	Act	Extruder3: Actual value screw torque	Analog	[%]	R
Hoff1_Force	Act	Haul-off 1: Actual value haul-off force	Analog	[kN]	R
Hoff2_Force	Act	Haul-off 2: Actual value haul-off force	Analog	[kN]	R
Ext1_Vacuum	Act	Ex1: Actual value vacuum	Analog	[mbar]	R
	MTol1	Ex1: -Tolerance vacuum	Analog	[mbar]	R / W
Ext2_Vacuum	Act	Ex2: Actual value vacuum	Analog	[mbar]	R
	MTol1	Ex2: -Tolerance vacuum	Analog	[mbar]	R / W
Ext3_Vacuum	Act	Ex3: Actual value vacuum	Analog	[mbar]	R
	MTol1	Ex3: -Tolerance vacuum	Analog	[mbar]	R / W
Ext1_MeltPress1	Act	Ex1: Actual value: melt pressure 1	Analog	[bar]	R
	Limit1	Ex1: Pre-warning melt pressure 1	Analog	[bar]	R / W
	Limit2	Ex1: Turning OFF by melt pressure 1	Analog	[bar]	R / W
Ext1_MeltPress2	Act	Ex1: Actual value: melt pressure 2	Analog	[bar]	R
	Limit1	Ex1: Pre-warning melt-pressure 2	Analog	[bar]	R / W
	Limit2	Ex1: Turning OFF by melt pressure 2	Analog	[bar]	R / W
Ext1_MeltPress3	Act	Ex1: Actual value: melt-pressure 3	Analog	[bar]	R
	Limit1	Ex1: Pre-warning melt-pressure 3	Analog	[bar]	R / W
	Limit2	Ex1: Turning OFF by melt-pressure 3	Analog	[bar]	R / W
Ext1_MeltPress4	Act	Ex1: Actual value: melt-pressure 4	Analog	[bar]	R
	Limit1	Ex1: Pre-warning melt-pressure 4	Analog	[bar]	R / W
	Limit2	Ex1: Turning OFF by melt-pressure 4	Analog	[bar]	R / W
Ext2_MeltPress1	Act	Ex2: Actual value: melt-pressure 1	Analog	[bar]	R
	Limit1	Ex2: Pre-warning melt-pressure 1	Analog	[bar]	R / W
	Limit2	Ex2: Turning OFF by melt pressure 1	Analog	[bar]	R / W
Ext3_MeltPress1	Act	Ex3: Actual value: melt-pressure 1	Analog	[bar]	R
	Limit1	Ex3: Pre-warning melt-pressure 1	Analog	[bar]	R / W
	Limit2	Ex3: Turning OFF by melt-pressure 1	Analog	[bar]	R / W
Gen_ReferenceMagnitude	Sel	Reference input during set value entries	Switch	0..2	R / W

4.7.2. Temperatures

Object	Element	Description	Type	Dim	Mode
Ext1_BarrelZone01	Set	Ex1: Set value temperature barrel zone1	Analog	[°C]	R / W
	Act	Ex1: Actual value temperature barrel zone1	Analog	[°C]	R
	PTol1	Ex1: +Tolerance barrel-zone 1	Analog	[°C]	R / W
	MTol1	Ex1: -Tolerance barrel-zone 1	Analog	[°C]	R / W
	XpH	Ex1: Proportional content heating barrel zone 1	Analog	[°/oo]	R / W
	XPC	Ex1: Proportional content cooling barrel zone 1	Analog	[°/oo]	R / W
	TNH	Ex1: Integral action time heating barrel zone 1	Analog	[s]	R / W
	TNC	Ex1: Integral action time cooling barrel zone 1	Analog	[s]	R / W
	TVH	Ex1: Derivative action time heating barrel zone 1	Analog	[s]	R / W
	TVK	Ex1: Derivative action time cooling barrel zone 1	Analog	[s]	R / W
	EDSet	Ex1: Set value ON-duration barrel zone 1	Analog	[%]	R / W
	EDAct	Ex1: Current heating-/cooling content barrel zone 1	Analog	[%]	R

Parameter objects for all temperatures are of the same structure. The tables above will therefore have to be used in an analogue manner for all subsequent temperature objects. With temperature zones that are not closed-loop controlled, all controller parameters are meaningless (from KPH downward). With heating zones that only heat or cool, the respective control parameters are without function.

Ext1_BarrelZone01	Ex1: Barrel zone 1
Ext1_BarrelZone02	Ex1: Barrel zone 2
Ext1_BarrelZone03	Ex1: Barrel zone 3
Ext1_BarrelZone04	Ex1: Barrel zone 4
Ext1_BarrelZone05	Ex1: Barrel zone 5
Ext1_BarrelZone06	Ex1: Barrel zone 6
Ext1_BarrelZone07	Ex1: Barrel zone 7
Ext1_BarrelZone08	Ex1: Barrel zone 8 (compounder only)
Ext1_BarrelZone09	Ex1: Barrel zone 9 (compounder only)
Ext1_BarrelZone10	Ex1: Barrel zone 10 (compounder only)
Ext1_BarrelZone11	Ex1: Barrel zone 11 (compounder only)
Ext1_BarrelZone12	Ex1: Barrel zone 12 (compounder only)
Ext1_ScrewTemp01	Ex1: Screw temperature control unit 1
Ext1_ScrewTemp02	Ex1: Screw temperature control unit 2
Ext1_AdapterZone01	Ex1: Adapter zone 1
Ext1_AdapterZone02	Ex1: Adapter zone 2
Ext1_AdapterZone03	Ex1: Adapter zone 3
Ext1_AdapterZone04	Ex1: Adapter zone 4
Ext1_MeltTemp01	Ex1: Melt temperature 1
Ext1_MeltTemp02	Ex1: Melt temperature 2
Ext1_BushZone01	Ex1: Grooved bushing temperature

Ext2_BarrelZone01	Ex2: Barrel zone 1
Ext2_BarrelZone02	Ex2: Barrel zone 2
Ext2_BarrelZone03	Ex2: Barrel zone 3
Ext2_BarrelZone04	Ex2: Barrel zone 4
Ext2_BarrelZone05	Ex2: Barrel zone 5
Ext2_BarrelZone06	Ex2: Barrel zone 6
Ext2_ScrewTemp01	Ex2: Screw temperature control unit 1
Ext2_ScrewTemp02	Ex2: Screw temperature control unit 2
Ext2_AdapterZone01	Ex2: Adapter zone 1
Ext2_MeltTemp01	Ex2: Melt temperature 1
Ext2_BushZone01	Ex2: Grooved bushing temperature

Ext3_BarrelZone01	Ex3: Barrel zone 1
Ext3_BarrelZone02	Ex3: Barrel zone 2
Ext3_BarrelZone03	Ex3: Barrel zone 3
Ext3_BarrelZone04	Ex3: Barrel zone 4

Ext3_BarrelZone05	Ex3: Barrel zone 5
Ext3_BarrelZone06	Ex3: Barrel zone 6
Ext3_ScrewTemp01	Ex3: Screw temperature control unit 1
Ext3_ScrewTemp02	Ex3: Screw temperature control unit 2
Ext3_AdapterZone01	Ex3: Adapter zone
Ext3_MeltTemp01	Ex3: Melt temperature 1
Ext3_BushZone01	Ex3: Grooved bushing temperature

Die_Zone01	Die temperature 1
Die_Zone02	Die temperature 2
Die_Zone03	Die temperature 3
Die_Zone04	Die temperature 4
Die_Zone05	Die temperature 5
Die_Zone06	Die temperature 6
Die_Zone07	Die temperature 7
Die_Zone08	Die temperature 8
Die_Zone09	Die temperature 9
Die_Zone10	Die temperature 10
Die_Zone11	Die temperature 11
Die_Zone12	Die temperature 12
Die_Zone13	Die temperature 13
Die_Zone14	Die temperature 14
Die_Zone15	Die temperature 15
Die_Zone16	Die temperature 16
Die_Zone17	Die temperature 17
Die_Zone18	Die temperature 18
Die_Zone19	Die temperature 19
Die_Zone20	Die temperature 20
Die_Zone21	Die temperature 21
Die_Zone22	Die temperature 22
Die_Zone23	Die temperature 23
Die_Zone24	Die temperature 24
Die_Zone25	Die temperature 25
Die_Zone26	Die temperature 26
Die_Zone27	Die temperature 27
Die_Zone28	Die temperature 28
Die_Zone29	Die temperature 29
Die_Zone30	Die temperature 30
Die_Zone31	Die temperature 31
Die_Zone32	Die temperature 32
Die_Zone33	Die temperature 33
Die_Zone34	Die temperature 34
Die_Zone35	Die temperature 35
Die_Zone36	Die temperature 36
Die_Zone37	Die temperature 37
Die_Zone38	Die temperature 38
Die_Zone39	Die temperature 39
Die_Zone40	Die temperature 40
Die_Zone41	Die temperature 41
Die_Zone42	Die temperature 42
Die_Zone43	Die temperature 43
Die_Zone44	Die temperature 44
Die_Zone45	Die temperature 45
Die_Zone46	Die temperature 46
Die_Zone47	Die temperature 47
Die_Zone48	Die temperature 48

4.7.3.Items of status information

Enquiries are made through the status information, which units are turned ON or OFF. 0 corresponds to the turned OFF status, and 1 means turned ON.

Object	Element	Description	Type	Dim	Mode
Ext1_ScrewSpeed	State	Ex1: Main drive turned ON	Switch	0..1	R
Ext2_ScrewSpeed	State	Ex2: Main drive turned ON	Switch	0..1	R
Ext3_ScrewSpeed	State	Ex3: Main drive turned ON	Switch	0..1	R
Ext1_Feed1_Speed	State	Extruder 1 feed unit 1 turned ON	Switch	0..1	R
Ext1_Feed2_Speed	State	Extruder 1 feed unit 2 turned ON	Switch	0..1	R
Ext1_Feed3_Speed	State	Extruder 1 feed unit 3 turned ON	Switch	0..1	R
Ext1_Feed4_Speed	State	Extruder 1 feed unit 4 turned ON	Switch	0..1	R
Ext2_Feed1_Speed	State	Extruder 2 feed unit 1 turned ON	Switch	0..1	R
Ext2_Feed2_Speed	State	Extruder 2 feed unit 2 turned ON	Switch	0..1	R
Ext2_Feed3_Speed	State	Extruder 2 feed unit 3 turned ON	Switch	0..1	R
Ext2_Feed4_Speed	State	Extruder 2 feed unit 4 turned ON	Switch	0..1	R
Hoff1_Speed	State	Haul-off 1 turned ON	Switch	0..1	R
Hoff2_Speed	State	Haul-off 2 turned ON	Switch	0..1	R
Ext1_BarrelHeating	State	Barrel heating turned ON	Switch	0..1	R
Ext1_ScrewHeating	State	Screw heat-balancing turned ON	Switch	0..1	R
Die_Heating	State	Die-heating turned ON	Switch	0..1	R
Gen_OperationMode	Sel	Current operating mode 0 = Off 1 = Service 3 = Production	Switch	0..1	R
Gen_LineState	Sel	Line and production status 0 = Stop 1 = Down 2 = Up 3 = Production running	Switch	0..3	R
Ext1_Vacuum1_State	Sel	Vacuum pump 1 Extruder 1 turned ON	Switch	0..1	R
Ext1_Vacuum2_State	Sel	Vacuum pump 2 Extruder 1 turned ON	Switch	0..1	R
Ext2_Vacuum1_State	Sel	Vacuum pump 1 Extruder 2 turned ON	Switch	0..1	R
Ext3_Vacuum1_State	Sel	Vacuum pump 1 Extruder 3 turned ON	Switch	0..1	R

4.7.4. Operating hours counter

Enquiries are made through the status information, which units are turned ON or OFF. 0 corresponds to the turned OFF status, and 1 means turned ON.

Object	Element	Description	Type	Dim	Mode
Ext1_OhcRunning	Span	Extruder 1 operating hours counter	Time	Sec	R
Ext2_OhcRunning	Span	Extruder 2 operating hours counter	Time	Sec	R
Ext3_OhcRunning	Span	Extruder 3 operating hours counter	Time	Sec	R
Ext4_OhcRunning	Span	Extruder 4 operating hours counter	Time	Sec	R
Ext5_OhcRunning	Span	Extruder 5 operating hours counter	Time	Sec	R
Ext1_Feed1_OhcRunning	Span	Operating hours counter for feed unit 1 of extruder 1	Time	Sec	R
Ext1_Feed2_OhcRunning	Span	Operating hours counter for feed unit 2 of extruder 1	Time	Sec	R
Ext1_Feed3_OhcRunning	Span	Operating hours counter for feed unit 3 of extruder 1	Time	Sec	R
Ext1_Feed4_OhcRunning	Span	Operating hours counter for feed unit 4 of extruder 1	Time	Sec	R
Ext1_Feed5_OhcRunning	Span	Operating hours counter for feed unit 5 of extruder 1	Time	Sec	R
Ext1_Feed6_OhcRunning	Span	Operating hours counter for feed unit 6 of extruder 1	Time	Sec	R
Ext2_Feed1_OhcRunning	Span	Operating hours counter for feed unit 1 of extruder 2	Time	Sec	R
Ext2_Feed2_OhcRunning	Span	Operating hours counter for feed unit 2 of extruder 2	Time	Sec	R
Ext2_Feed3_OhcRunning	Span	Operating hours counter for feed unit 3 of extruder 2	Time	Sec	R
Ext2_Feed4_OhcRunning	Span	Operating hours counter for feed unit 4 of extruder 2	Time	Sec	R
Ext3_Feed1_OhcRunning	Span	Operating hours counter for feed unit 1 of extruder 3	Time	Sec	R
Ext3_Feed2_OhcRunning	Span	Operating hours counter for feed unit 2 of extruder 3	Time	Sec	R
Ext3_Feed3_OhcRunning	Span	Operating hours counter for feed unit 3 of extruder 3	Time	Sec	R
Ext3_Feed4_OhcRunning	Span	Operating hours counter for feed unit 4 of extruder 3	Time	Sec	R
ChillRoll1_OhcRunning	Span	Cooling roll 1 operating hours counter	Time	Sec	R
ChillRoll2_OhcRunning	Span	Cooling roll 2 operating hours counter	Time	Sec	R
EmbRoll1_OhcRunning	Span	Embossing roller 1 operating hours counter	Time	Sec	R
PressRoll1_OhcRunning	Span	Press roll 1 operating hours counter	Time	Sec	R
PressRoll2_OhcRunning	Span	Press roll 2 operating hours counter	Time	Sec	R
PolRoll1_OhcRunning	Span	Bottom polishing roll operating hours counter	Time	Sec	R
PolRoll2_OhcRunning	Span	Middle polishing roll operating hours counter	Time	Sec	R
PolRoll3_OhcRunning	Span	Top polishing roll operating hours counter	Time	Sec	R
Hoff1_OhcRunning	Span	Haul-off 1 operating hours counter	Time	Sec	R
Hoff2_OhcRunning	Span	Haul-off 2 operating hours counter	Time	Sec	R

4.7.5. Seven-day timer

Object	Element	Description	Type	Dim	Mode
Gen_WeekTimerOn	Sel	Activate seven-day timer	Switch	0: OFF 1: ON	R / W
Gen_Week_TimerExt1Bar	Monday	Switching time Monday, barrel heater for extruder 1	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, barrel heater for extruder 1	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, barrel heater for extruder 1	Time	hh:mm	R / W
	Thursday	Switching time Thursday, barrel heater for extruder 1	Time	hh:mm	R / W
	Friday	Switching time Friday, barrel heater for extruder 1	Time	hh:mm	R / W
	Saturday	Switching time Saturday, barrel heater for extruder 1	Time	hh:mm	R / W
	Sunday	Switching time Sunday, barrel heater for extruder 1	Time	hh:mm	R / W
Gen_Week_TimerExt2Bar	Monday	Switching time Monday, barrel heater for extruder 2	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, barrel heater for extruder 2	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, barrel heater for extruder 2	Time	hh:mm	R / W
	Thursday	Switching time Thursday, barrel heater for extruder 2	Time	hh:mm	R / W
	Friday	Switching time Friday, barrel heater for extruder 2	Time	hh:mm	R / W
	Saturday	Switching time Saturday, barrel heater for extruder 2	Time	hh:mm	R / W
	Sunday	Switching time Sunday, barrel heater for extruder 2	Time	hh:mm	R / W
Gen_Week_TimerExt3Bar	Monday	Switching time Monday, barrel heater for extruder 3	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, barrel heater for extruder 3	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, barrel heater for extruder 3	Time	hh:mm	R / W
	Thursday	Switching time Thursday, barrel heater for extruder 3	Time	hh:mm	R / W
	Friday	Switching time Friday, barrel heater for extruder 3	Time	hh:mm	R / W
	Saturday	Switching time Saturday, barrel heater for extruder 3	Time	hh:mm	R / W
	Sunday	Switching time Sunday, barrel heater for extruder 3	Time	hh:mm	R / W
Gen_Week_TimerExt1Scr	Monday	Switching time Monday, screw heater for extruder 1	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, screw heater extruder 1	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, screw heater for extruder 1	Time	hh:mm	R / W
	Thursday	Switching time Thursday, screw heater for extruder 1	Time	hh:mm	R / W
	Friday	Switching time Friday, screw heater for extruder 1	Time	hh:mm	R / W
	Saturday	Switching time Saturday, screw heater for extruder 1	Time	hh:mm	R / W
	Sunday	Switching time Sunday, screw heater for extruder 1	Time	hh:mm	R / W
Gen_Week_TimerExt2Scr	Monday	Switching time Monday, screw heater for extruder 2	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, screw heater extruder 2	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, screw heater for extruder 2	Time	hh:mm	R / W
	Thursday	Switching time Thursday, screw heater for extruder 2	Time	hh:mm	R / W
	Friday	Switching time Friday, screw heater for extruder 2	Time	hh:mm	R / W
	Saturday	Switching time Saturday, screw heater for extruder 2	Time	hh:mm	R / W
	Sunday	Switching time Sunday, screw heater for extruder 2	Time	hh:mm	R / W
Gen_Week_TimerExt3Scr	Monday	Switching time Monday, screw heater for extruder 3	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, screw heater extruder 3	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, screw heater for extruder 3	Time	hh:mm	R / W
	Thursday	Switching time Thursday, screw heater for extruder 3	Time	hh:mm	R / W
	Friday	Switching time Friday, screw heater for extruder 3	Time	hh:mm	R / W
	Saturday	Switching time Saturday, screw heater for extruder 3	Time	hh:mm	R / W
	Sunday	Switching time Sunday, screw heater for extruder 3	Time	hh:mm	R / W
Gen_WeekTimerDie	Monday	Switching time Monday, die heater	Time	hh:mm	R / W
	Tuesday	Switching time Tuesday, die heater	Time	hh:mm	R / W
	Wednesday	Switching time Wednesday, die heater	Time	hh:mm	R / W
	Thursday	Switching time Thursday, die heater	Time	hh:mm	R / W
	Friday	Switching time Friday, die heater	Time	hh:mm	R / W
	Saturday	Switching time Saturday, die heater	Time	hh:mm	R / W
	Sunday	Switching time Sunday, die heater	Time	hh:mm	R / W

4.7.7. Saw-cut control and pipe-wall thickening (option)

Object	Element	Description	Type	Dim	Mode
Saw1_ThickPipeType	Sel	Wall-thickening mode	Switch	0: OFF 1: front 2: rear 3: bilateral 4: alternating	R / W
Saw1_Lot1	SetThickLen	Length of thickening	Analog	[m]	R / W
Saw1_Lot1	SetSpeedRed	Reduction in haul-off speed	Analog	[%]	R / W
Saw1_ThickDistance	Set	Distance from pipe head to saw blade	Analog	[%]	R / W
Saw1_BladeThickness	Set	Saw-blade thickness	Analog	[mm]	R / W
Saw1_MinimumLength	Set	Minimum saw cut length	Analog	[m]	R / W
Saw1_Shrinkage	Set	Shrinkage factor	Analog	[%]	R / W
Saw1_State	Sel	Activate saw-cut control	Switch	0: OFF 1: ON	R / W
Saw1_MonitorTimer	Time	Last saw cutting time	Analog	[s]	R
Saw1_TotalBadPcs	Set	Fail pieces total	Analog	[Qty]	R
Saw1_TotalBadLen	Act	Waste length total	Analog	[m]	R
Saw1_TotalGoodPcs	Set	Sum total of good pieces	Analog	[Qty]	R
Saw1_TotalGoodLen	Act	Sum total of good lengths	Analog	[m]	R
Saw1_TipperTime	Time	Activation time for tipper	Analog	[%]	R / W
Saw1_TipperDistance	Set	Distance of saw blade to front edge of tipper	Analog	[m]	R / W
Saw1_TipperSelect	Sel	Tipper function ON	Switch	[0,1]	R / W
Saw1_Lot1	SetCutLen	Cut lengths in lot 1	Analog	[m]	R
Saw1_Lot1	ProdLen	Product lengths in lot 1	Analog	[m]	R / W
Saw1_Lot1	SetSeqCnt	Set value repeat number in lot 1	Analog	[Qty]	R / W
Saw1_Lot1	ActSeqCnt	Actual value repeat number in lot 1	Analog	[Qty]	R
Saw1_Lot1	SetGoodLen	Set lengths in lot 1	Analog	[m]	R / W
Saw1_Lot1	SetGoodPcs	Set pieces in lot 1	Analog	[m]	R / W
Saw1_Lot1	ActBadLen	Actual reject lengths in lot 1	Analog	[m]	R
Saw1_Lot1	ActGoodLen	Actual good lengths in lot 1	Analog	[m]	R
Saw1_Lot1	ActBadPcs	Actual reject pieces in lot 1	Analog	[Qty]	R
Saw1_Lot1	ActGoodPcs	Actual good pieces in lot 1	Analog	[Qty]	R
Saw1_Lot2	SetCutLen	Cut lengths in lot 2	Analog	[m]	R
Saw1_Lot2	ProdLen	Product lengths in lot 2	Analog	[m]	R / W
Saw1_Lot2	SetSeqCnt	Set value repeat number in lot 2	Analog	[Qty]	R / W
Saw1_Lot2	ActSeqCnt	Actual value repeat number in lot 2	Analog	[Qty]	R
Saw1_Lot2	SetGoodLen	Set lengths in lot 2	Analog	[m]	R / W
Saw1_Lot2	SetGoodPcs	Set pieces in lot 2	Analog	[m]	R / W
Saw1_Lot2	ActBadLen	Actual reject lengths in lot 2	Analog	[m]	R
Saw1_Lot2	ActGoodLen	Actual good lengths in lot 2	Analog	[m]	R
Saw1_Lot2	ActBadPcs	Actual reject pieces in lot 2	Analog	[Qty]	R
Saw1_Lot2	ActGoodPcs	Actual good pieces in lot 2	Analog	[Qty]	R
...					
Saw1_Lot5	SetCutLen	Cut lengths in lot 5	Analog	[m]	R
Saw1_Lot5	ProdLen	Product lengths in lot 5	Analog	[m]	R / W
Saw1_Lot5	SetSeqCnt	Set value repeat number in lot 5	Analog	[Qty]	R / W
Saw1_Lot5	ActSeqCnt	Actual value repeat number in lot 5	Analog	[Qty]	R
Saw1_Lot5	SetGoodLen	Set lengths in lot 5	Analog	[m]	R / W
Saw1_Lot5	SetGoodPcs	Set pieces in lot 5	Analog	[m]	R / W
Saw1_Lot5	ActBadLen	Actual reject lengths in lot 5	Analog	[m]	R
Saw1_Lot5	ActGoodLen	Actual good lengths in lot 5	Analog	[m]	R
Saw1_Lot5	ActBadPcs	Actual reject pieces in lot 5	Analog	[Qty]	R
Saw1_Lot5	ActGoodPcs	Actual good pieces in lot 5	Analog	[Qty]	R

If the saw cut control is operated in conjunction with dual-rope extrusion, all listed parameters are doubled. All parameters referring to the second strand begin with "Saw2" instead of "Saw1" in the object name.

4.7.8. Ultrasonic wall-thickness gauging (option)



Ultrasonic parameters can only be set or read, when the US-measuring system has been connected to the C5 and is operational.

Object	Element	Description	Type	Dim	Mode
Us1_Measure	Sel	Ultrasonic measurement	Switch	0: OFF 1: ON	R / W
Us1_Thickness	Set	Set value wall-thickness	Analog	[mm]	R / W
	Act	Actual average wall-thickness	Analog	[mm]	R
	Min	Actual minimum wall thickness	Analog	[mm]	R
	Max	Actual maximum wall thickness	Analog	[mm]	R
	Limit1	Upper wall thickness limiting value	Analog	[mm]	R / W
	Limit2	Lower wall thickness limiting value	Analog	[mm]	R / W
Us1_Diameter	Limit4	Correction value for wall thickness	Analog	[%]	R / W
	Set	Set value pipe diameter	Analog	[mm]	R / W
	Act	Actual average diameter	Analog	[mm]	R
	Min	Actual minimum diameter	Analog	[mm]	R
	Max	Actual maximum diameter	Analog	[mm]	R
	Limit1	Upper diameter limiting value	Analog	[mm]	R / W
Us1_Excentricity	Limit2	Lower diameter limiting value	Analog	[mm]	R / W
	Limit4	Correction value for diameter	Analog	[mm]	R / W
	Max	Specified maximum eccentricity	Analog	[mm]	R / W
Us1_Ovality	Act	Actual eccentricity	Analog	[mm]	R
	Max	Specified maximum ovality	Analog	[mm]	R / W
Us1_AutoCalibration	Act	Actual ovality	Analog	[mm]	R
	Sel	Automatic calibration	Switch	0: OFF 1: ON	R / W
Us1_SminControl	Sel	Thin-spot regulation	Switch	0: OFF 1: ON	R / W
Us1_SectorThickness	Val1	Wall-thickness value 1 (at 0°)	Analog	[mm]	R
Us1_SectorThickness	Val2	Wall-thickness value 1 (at 45°)	Analog	[mm]	R
Us1_SectorThickness	Val3	Wall-thickness value 1 (at 90°)	Analog	[mm]	R
Us1_SectorThickness	Val4	Wall-thickness value 1 (at 135°)	Analog	[mm]	R
Us1_SectorThickness	Val5	Wall-thickness value 1 (at 180°)	Analog	[mm]	R
Us1_SectorThickness	Val6	Wall-thickness value 1 (at 225°)	Analog	[mm]	R
Us1_SectorThickness	Val7	Wall-thickness value 1 (at 270°)	Analog	[mm]	R
Us1_SectorThickness	Val8	Wall-thickness value 1 (at 315°)	Analog	[mm]	R
Us1_SpecificWeight	Set	Specific weight of the material	Analog	[g/cm³]	R / W
Us1_DistanceDieScanner	Set	Distance from pipe head to US scanner	Analog	[m]	R / W

In case of a dual strand extrusion with 2 ultrasonic systems, all of the parameters are also available for the second ultrasonic system. For all parameters referring to the second US system, the object name begins with "Us2" instead of "Us1".

4.7.9. Downstream for sheet lines (option)

Object	Element	Description	Type	Dim	Mode
PolRoll1_Speed	Set	Set value bottom roller speed	Analog	[m/min]	R / W
	Act	Actual value bottom roller speed	Analog	[m/min]	R
PolRoll2_Speed	Set	Set value middle roller speed	Analog	[m/min]	R / W
	Act	Actual value middle roller speed	Analog	[m/min]	R
PolRoll3_Speed	Set	Set value top roller speed	Analog	[m/min]	R / W
	Act	Actual value top roller speed	Analog	[m/min]	R
EmbRoll1_Speed	Set	Set value embossing roller speed	Analog	[m/min]	R / W
	Act	Actual value embossing roller speed	Analog	[m/min]	R
Hoff1_Speed	Set	Set value haul-off speed	Analog	[m/min]	R / W
	Act	Actual value haul-off speed 1	Analog	[m/min]	R
Hoff2_Speed	Act	Actual value haul-off speed 2	Analog	[m/min]	R
ChillRoll1_Speed	Set	Set value upper cooling roller 1 speed	Analog	[m/min]	R / W
	Act	Actual value upper cooling roller 1 speed	Analog	[m/min]	R
ChillRoll2_Speed	Set	Set value lower cooling roller 1 speed	Analog	[m/min]	R / W
	Act	Actual value lower cooling roller 1 speed	Analog	[m/min]	R
PolRoll1_Temp	Set	Ex1: Set value temperature bottom roller	Analog	[°C]	R / W
	Act	Ex1: Actual value temperature bottom roller	Analog	[°C]	R
	PTol1	Ex1: +tolerance temperature bottom roller	Analog	[°C]	R / W
	MTol1	Ex1: -tolerance temperature bottom roller	Analog	[°C]	R / W
PolRoll2_Temp	EDAct	Ex1: Current heating-/cooling content bottom roller	Analog	[%]	R
	Set	Ex1: Set value temperature middle roller	Analog	[°C]	R / W
	Act	Ex1: Actual value temperature middle roller	Analog	[°C]	R
	PTol1	Ex1: + tolerance temperature middle roller	Analog	[°C]	R / W
PolRoll3_Temp	MTol1	Ex1: - tolerance temperature middle roller	Analog	[°C]	R / W
	EDAct	Ex1: Current heating-/cooling content middle roller	Analog	[%]	R
	Set	Ex1: Set value temperature top roller	Analog	[°C]	R / W
	Act	Ex1: Actual value temperature top roller	Analog	[°C]	R
EmbRoll1_Temp	PTol1	Ex1: +tolerance temperature top roller	Analog	[°C]	R / W
	MTol1	Ex1: - tolerance temperature top roller	Analog	[°C]	R / W
	EDAct	Ex1: Current heating-/cooling content top roller	Analog	[%]	R
	Set	Ex1: Set value temperature embossing roller	Analog	[°C]	R / W
PolRoll1_Speed	Act	Ex1: Actual value temperature embossing roller	Analog	[°C]	R
	PTol1	Ex1: +tolerance temperature embossing roller	Analog	[°C]	R / W
	MTol1	Ex1: - tolerance temperature embossing roller	Analog	[°C]	R / W
	EDAct	Ex1: Current heating-/cooling content embossing roller	Analog	[%]	R
PolRoll1_Speed	State	Bottom roller switched on	Switch	0..1	R
PolRoll2_Speed	State	Middle roller switched on	Switch	0..1	R
PolRoll3_Speed	State	Top roller switched on	Switch	0..1	R
ChillRoll1_Speed	State	Foam cooling roll 1 turned ON	Switch	0..1	R
ChillRoll2_Speed	State	Foam cooling roll 2 turned ON	Switch	0..1	R
EmbRoll1_Speed	State	Embossing roller 1 turned ON	Switch	0..1	R
Sheet_GapTopLeft	Act	Roller gap top left	Analog	[mm]	R
Sheet_GapTopRight	Act	Roller gap top right	Analog	[mm]	R
Sheet_GapBottomLeft	Act	Roller gap bottom left	Analog	[mm]	R
Sheet_GapBottomRight	Act	Roller gap bottom right	Analog	[mm]	R

4.7.10. Melt pump (option)

Object	Element	Description	Type	Dim	Mode
MeltPump1_Speed	Set	Set value melt pump speed	Analog	[rpm]	R / W
	Act	Actual value melt pump speed	Analog	[rpm]	R
MeltPump1_MinSpeed	Set	Minimum speed melt pump	Analog	[rpm]	R / W
MeltPump1_MinPressure	Set	Melt press limit melt pump without material	Analog	[bar]	R / W
MeltPump1_IdleTime	Time	Monitoring time melt pump without material	Analog	[s]	R / W
MeltPump1_Torque	Act	Actual value melt pump torque	Analog	[%]	R
MeltPump1_Speed	State	Melt pump switched on	Switch	0..1	R
PnCtrl1_CtrlMode	Sel	P/N control mode	Switch	0..pump 1..extruder	R / W
PnCtrl1_State	Sel	P/N-Control switched on	Switch	0..1	R
PnCtrl1_Pid	Ta	Cycle time P/N-Control	Analog	[s]	R / W
	Tn	Control parameter: I-value P/N-Control	Analog	[s]	R / W
	Tv	Control parameter: D-value P/N-Control	Analog	[s]	R / W
	Xp	Control parameter: P-value P/N-Control	Analog	[°/oo]	R / W

5. Programming a host interface

5.1. General notes

The interface for linking up a C5 machine can be programmed on all systems or platforms which provide and support the interfaces described in Chapter 2 (TCP/IP via 10BaseT or 100BaseT Ethernet) using suitable drivers.

The C5 computer interface uses a master-slave protocol, i.e. the C5 (slave) responds only when the host computer (master) sends a telegram. The host computer must wait until it receives a response from the C5. The C5 responds to every telegram, even if only with an error message. If no answer can be received, the connection to the C5 must be checked (correct baud rate, correct cable connection, correct network address, etc.).

5.2. Connection setup

Before the first connection setup, an I command should always be performed. This returns the C5 computer interface to its initial state and deletes all parameter definitions.



As long as initialisation of the C5 control system has not been completed, no connection can be set up via the C5 computer interface. This is shown by the error message "?" received in response to an I command. The I command should therefore be repeated until a positive acknowledgement ("I:*") is received from the C5.

After the I command, it is advisable to request a handle (Q command) for all parameters which will be read or written frequently via the interface. The use of handles reduces the quantity of data on the interface and simplifies the management of the parameters. The handle is a 4-digit unambiguous number which can be used instead of the (relatively long) parameter name as long as the connection is maintained. In addition the handles are assigned with ascending numbers, so that they can be used at the same time as indexes in a management list for the parameters.

Example:

Inquiry from the host

Q:Ext1_Speed.Set.VAL|Ext1_Speed.Set.VAL|Ext1_Feed1_Speed.Set.VAL|Ext1_Feed1_Speed.Act.VAL

Response from the C5

Q:\$0000|\$0001|\$0002|\$0003

Inquiry from the host

Q:Ext1_BarrelZone01.Act.VAL|Ext1_BarrelZone02.Act.VAL|Ext1_BarrelZone03.Act.VAL

Response from the C5

Q: \$0004|\$0005|\$0006

etc.

Thus it is possible to draw up a table containing all the information required concerning each parameter.

Name (string[16])	Value (string[20])	Type of parameter (WORD)	...
Ext1_Speed.Set.VAL	20.5	decimal	
Ext1_Speed.Act.VAL	20.4	decimal	
Ext1_Feed1_Speed.Set.VAL	122.1	decimal	
Ext1_Feed1_Speed.Act.VAL	122.0	decimal	
Ext1_BarrelZone01.Act.VAL	187	decimal	
Ext1_BarrelZone02.Act.VAL	188	decimal	
Ext1_BarrelZone03.Act.VAL	185	decimal	
etc.			

By means of the handle (NOTE: the handle is a hexadecimal number) it is possible to directly consult the table and have access to all the information concerning a parameter.

After drawing up the table, all current parameter values must be read out (R command) and entered in the table (in the value field).

Finally all parameters which can change their value during the interface connection should be included in a request list (S command). The listing of a parameter causes the C5 control

system to register changes in the value of this parameter and to transmit them to the host upon request (using the N command).

Static information concerning the machine (e.g. version or machine name) does not need to be included in the list. It is sufficient if these data are read once after the connection has been set up.

5.3.Data exchange

In general, parameter values should only be sent to the C5 when they have changed, in order to avoid unnecessary load on the interface.

The parameter values of the C5 must be read out by polling (cyclic readout), since the C5 computer interface only reacts to requests from the host system and thus cannot send information concerning changes in parameter values of its own accord. However, the polling procedure can be reduced to a minimum by using a parameter request list (see 'S' command). Listing parameters amounts to telling the C5 that information will be required concerning any changes in certain parameters. A single command (see 'N' command) suffices to obtain a response telegram showing all changed parameters together with the new value. If several parameters have changed, they must be read out by repeating the N command.

After the connection setup has been completed (see chapter 6.2) it suffices to poll parameter changes using the N command.

If information is also required concerning the alarm status of the machine, it is possible to read out the current alarms of the C5 by cyclic repeat performance of the A command. First the number of alarms should be requested using the command A:COUNT. If the number has changed since the last inquiry, details of each alarm can be read out using a further A command.

5.4. Severed connection

If no response is received after sending a telegram, this is a sign that the connection has been severed. In all cases the maximum response time should be allowed to pass before taking further measures.

With TP or HP, one or two repeat attempts should be made following a waiting period, before declaring the connection severed.

With NTP, error-free transmission is ensured by TCP/IP. Therefore no repetition of the telegram is necessary for network connections.

In general, following a severed connection, the receive and send buffers and all variables or tables relating to the control of the connection should be deleted or reset. Then a new connection setup should be started, beginning with the I command.

5.5. Programming a TCP/IP interface

The program examples below show in principle the procedure for connecting to the C5.

The sample codes contained in the examples have not been checked for errors and serve as guidelines only.

In any case, the codes must be adapted to the operating system used and the application structure.

5.5.1.Example program in C

```

/*****
/*
/*   C example file for C5 network connection
/*
/*****

#include <stdio.h>
#include <stdlib.h>
#include <types.h>
#include <netinet/in.h>
#include <sys/socket.h>
#include <netdb.h>

main(int argc, char *argv[])
{
    unsigned short port;          /* port client will connect to          */
    char buf[1024];               /* data buffer for sending and receiving */
    char address[32];             /* server address                      */
    struct sockaddr_in server;    /* server address                      */
    int s;                       /* client socket                      */

    /*
     * Initialize with sockets.
     */
    sock_init();

    /*
     * Get the C5 address.
     */
    strcpy(address, argv[1]);
    if (strlen(address) == 0)
    {
        fprintf(stderr, "Invalid C5 address\n");
        exit(2);
    }

    /*
     * port number for C5 central computer interface
     */
    port = 18901;

    /*
     * Put the server information into the server structure.
     * The port must be put into network byte order.
     */
    server.sin_family      = AF_INET;
    server.sin_port        = htons(port);
    server.sin_addr.s_addr = inet_addr(address);

    /*
     * Get a stream socket.
     */

```

```

if ((s = socket(PF_INET, SOCK_STREAM, 0)) < 0)
{
    psock_errno("Socket()");
    exit(3);
}

/*
 * Connect to the server.
 */
if (connect(s, (struct sockaddr *)&server, sizeof(server)) < 0)
{
    psock_errno("Connect()");
    exit(4);
}

/*
 * LOGIN to C5
 */
buf[0] = 1;                ; data type
buf[1] = 0;                ; high order byte of data length
buf[2] = 0;                ; low order byte of data length

if (send(s, buf, 3, 0) < 0)
{
    psock_errno("Send(LOGIN)");
    exit(5);
}

/*
 * Put the INIT command message into the buffer.
 */
buf[0] = 0;                ; data type
buf[1] = 0;                ; high order byte of data length
buf[2] = 2;                ; low order byte of data length
buf[3] = 'I';              ; INIT command
buf[4] = ':';

if (send(s, buf, 5, 0) < 0)
{
    psock_errno("Send(INIT)");
    exit(5);
}

/*
 * The C5 sends back the answer. Receive it into the buffer.
 */
if (recv(s, buf, sizeof(buf), 0) < 0)
{
    psock_errno("Recv()");
    exit(6);
}

/*
 * Close the socket.
 */
soclose(s);

exit(0);
}

```

Example:

TEST 197.136.153.20

5.5.2. .NET example program in C#

```

using System;
using System.Threading;
using System.Net.Sockets;
using System.Net;

namespace C5Connection
{
    /// <summary>
    /// Class for C5 Central Computer Interface Client
    /// </summary>
    public class C5Client
    {
        private TcpClient m_client;
        private NetworkStream m_stream;

        /// <summary>Constructor</summary>
        public C5Client()
        {
            m_client = new TcpClient();
            m_client.Connect(IPAddress.Parse("196.100.0.17"), 18901);

            // Open Data stream
            m_stream = m_client.GetStream();

            // Send Login
            SendLogin();

            // Send INIT Command
            SendCommand("I:X");

            // Wait until any data is available
            while (!m_stream.DataAvailable)
                ;

            // Read the answer
            string s = ReceiveData();
            Debug.WriteLine("Received string: " + s);
        }

        private string ReceiveData()
        {
            byte[] ab = new byte[8192];
            int nRead = 0;

            // Read the first three bytes (header)
            do
            {
                nRead += m_stream.Read(ab, nRead, 1);
            } while (m_stream.DataAvailable && (nRead < 3));

            // Check header size
            if (nRead != 3)
            {
                // Error: Header is too short
                return String.Empty;
            }

            // Examine Data Length
            int len = (int)ab[1] << 8 | (int)ab[2];

            // Read data
            m_stream.Read(ab, 3, len);
        }
    }
}

```

```
        // Convert bytes to string
        return Encoding.ASCII.GetString(ab, 3, len);
    }

    private void SendLogin()
    {
        byte[] ab = new byte[8192];
        ab[0] = 1; // Data Type = Login
        ab[1] = 0;
        ab[2] = 0;

        m_stream.Write(ab, 0, 3);
    }

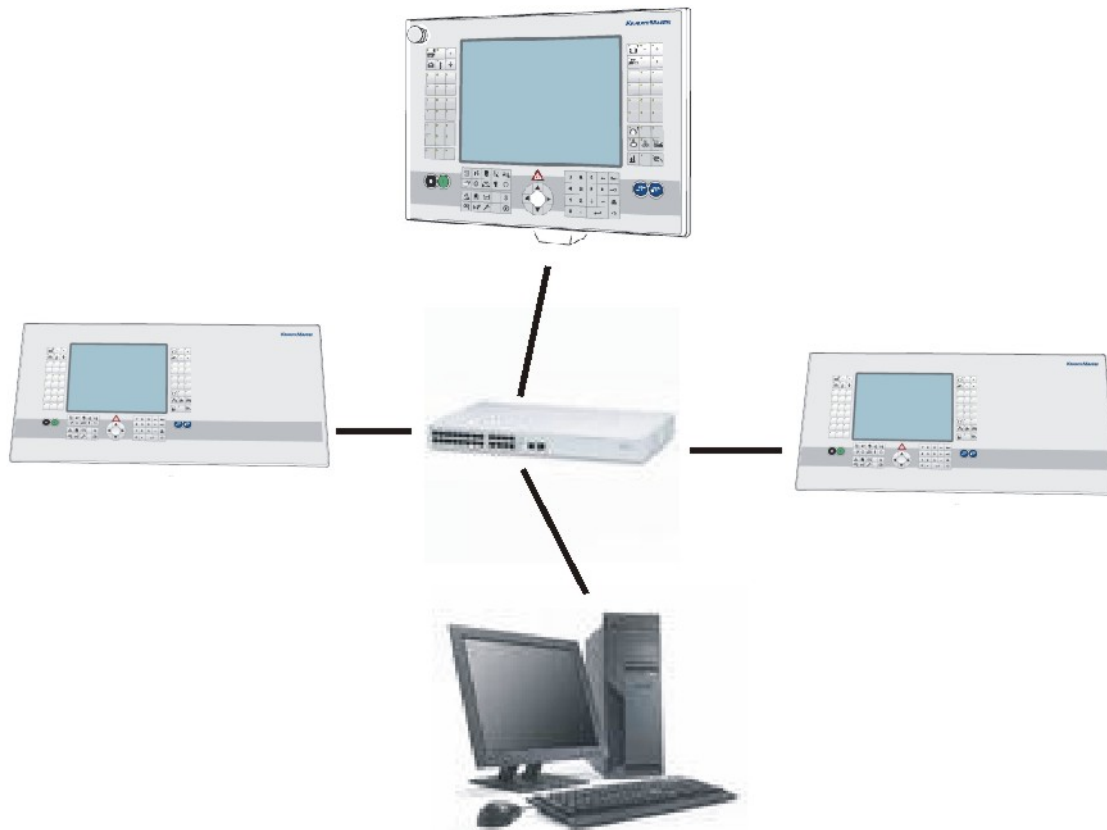
    private void SendCommand(string cmd)
    {
        byte[] ab = new byte[8192];
        ab[0] = 0; // Data Type = Command
        ab[1] = (byte)(cmd.Length >> 8);
        ab[2] = (byte)(cmd.Length & 0xFF);
        Encoding.ASCII.GetBytes(cmd, 0, cmd.Length, ab, 3);

        m_stream.Write(ab, 0, cmd.Length + 3);
    }
}
```

6. Establishing the network connection

6.1.Overview

For the network connection, the machine (Ethernet2) as well as the remote PC must be equipped with a network adapter.



An **Ethernet** network of type **10/100BaseT** is supported. This is a 10/100 MBit Ethernet with Twisted Pair cables and with RJ45 connecting plugs (Western plugs) at both ends. The network has a **star-shaped** structure, and each machine must have a connection to the **hub/switch** (star distributor). The hub/switch forms the central exchange.

6.2. Wiring

Twisted Pair cables are used as connecting cables. The length of the cables from and to the hub must not exceed **100 meters**. To help prevent noise and interference, STP cables (Shielded Twisted Pair) can be used instead of UTP cables (Unshielded Twisted Pair).



Assignment and cable colour codes for RJ45 plugs

Wire pair	Pins	EIA/TIA	IEC	REA	DIN 47.100
1	4/5	blue/white	white/blue	white/blue	white/brown
2	3/6	white/orange	red/orange	turquoise/violet	green/yellow
3	1/2	white/green	black/grey	white/orange	grey/pink
4	7/8	white/brown	yellow/brown	turquoise/violet	blue/red

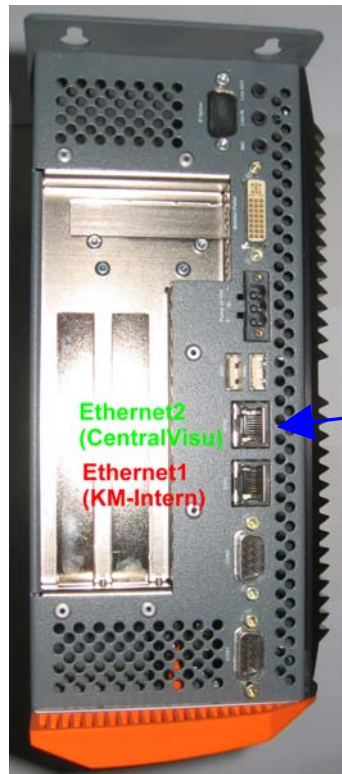
Assignment of RJ45 plugs for Ethernet (network cable)

6.3. Connection to the C5

6.3.1. Cable connection

The network connection on the C5 is located on the top of the visualisation PC. This computer has 2 network adapters. Use the **ETH2** connection.

ETH1 is available for internal communication by Krauss Maffei only.



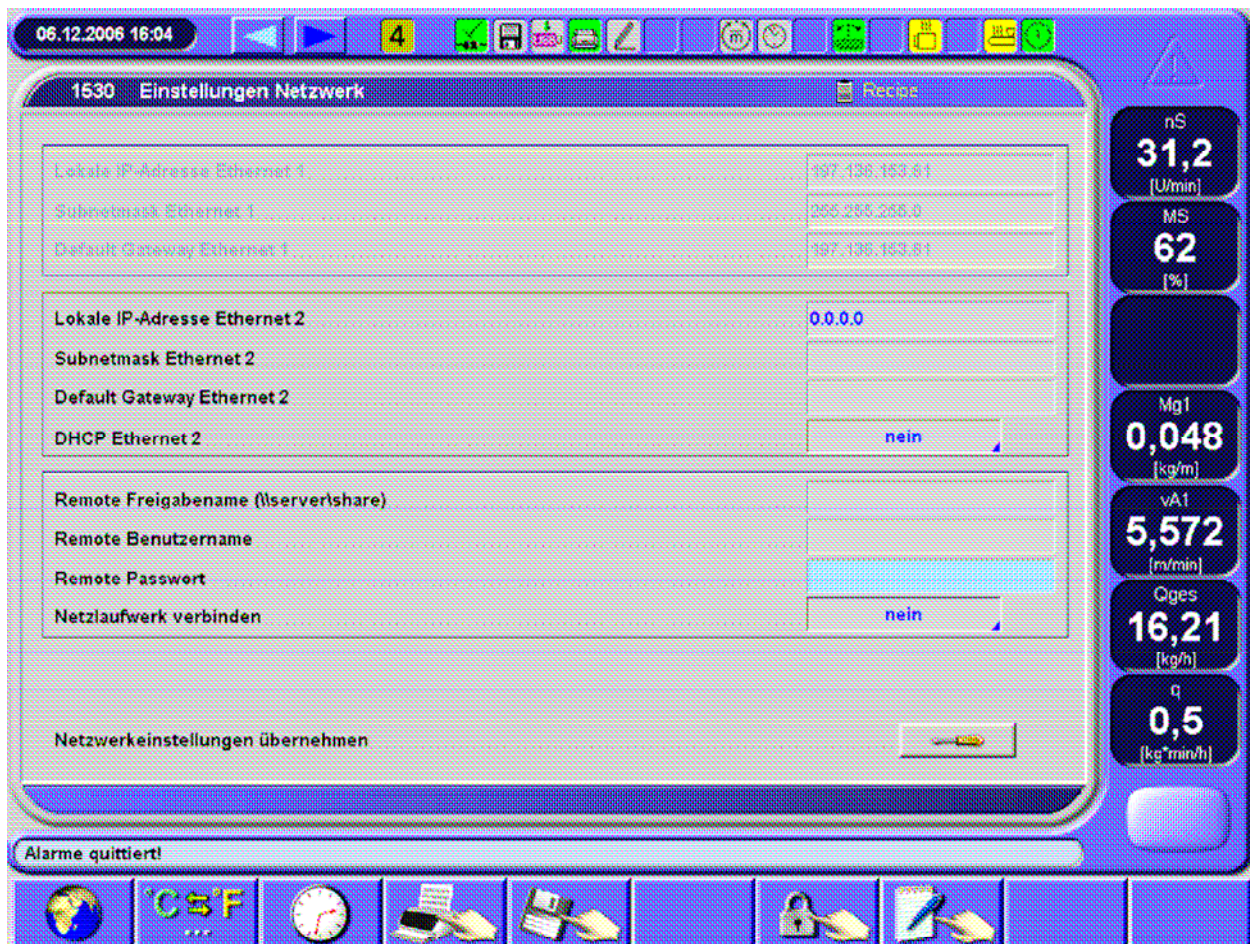
6.3.2. Network configuration

Press the following buttons to open the network settings page:



, then

The connection settings can then be configured in access level 4



For the network connection, ensure that different addresses have been programmed for each network station. The standard address for connection to C5 machines can be determined by the system administrator (for example: 196.136.153.xxx -> xxx: number between 1 and 254). For the central visualisation, the setting for "DHCP Ethernet 2" must always be "no". Otherwise, no clear machine assignment is possible.

IMPORTANT: Settings configured on this page will not be accepted until the **Apply network settings** button on the lower edge of the screen is pressed. To save the settings permanently, it is imperative that a **visualisation warm start** is actuated on page 1765.



The network address is not visible until a connection to the destination has been established (i.e. C5 extruder is connected to a switch)

6.4. Network hub / switch

The network has a **star-shaped** structure, and each machine must have a connection to the **hub** (star distributor). The hub forms the central exchange. There is a line from the hub to the remote PC. If you are using a switch instead of a hub, be absolutely sure that the switch has been configured correctly for the respective addresses.

The networking of machines and remote computers should be carried out by a technician.

6.5. Connection to remote computer

The remote computer must be equipped with a 100mbit Ethernet network adapter. There are a number of manufacturers who supply suitable plug-in cards. Ensure that drivers are available which are compatible with the operating system used.

6.5.1. Network installation

In order to operate the Ethernet network, the driver for the network adapter and the TCP/IP protocol must be installed on the remote PC and a link must be established between adapter and protocol.

If you have Windows, this is found when configuring the settings by going to:

Start menu → Settings → Network Connections → LAN connection X

Attention:

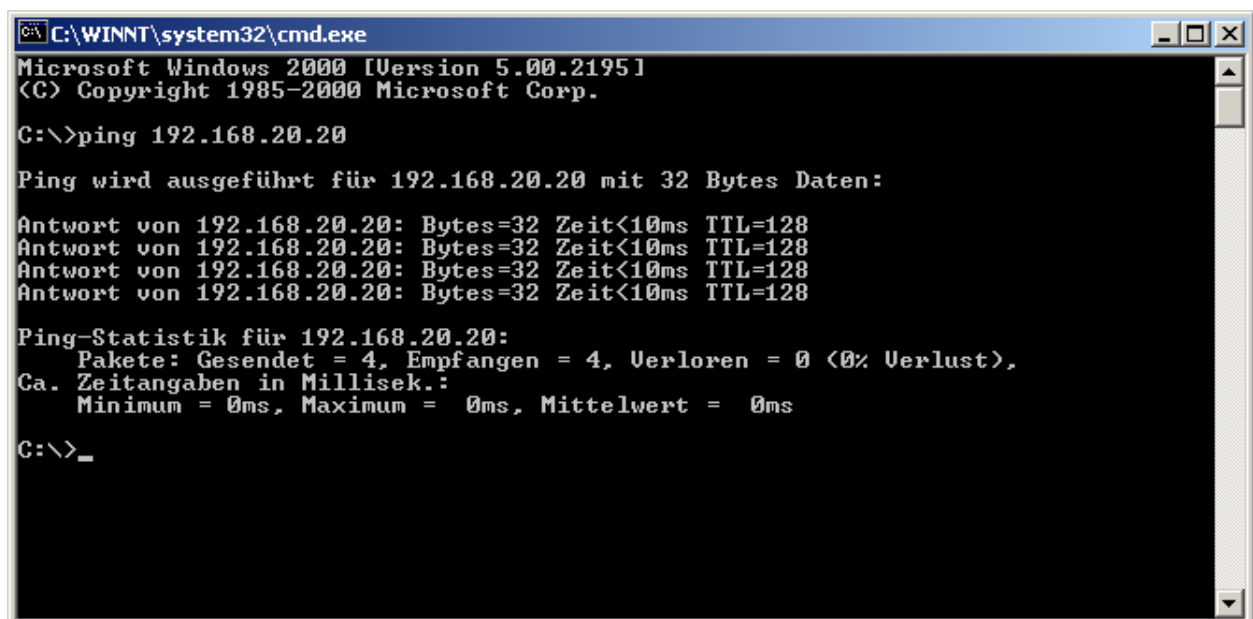
1. The network interface of the master computer must be on the same network as the extruder that is to be monitored.
2. Assign a static network address to the master computer and do not use DHCP.

6.6. Testing the network connection

Use the ping command to determine whether a specific network address is accessible. Under **Start menu → Run...**, open the command line editor by entering **cmd**.

Use the command `ping 192.168.20.20`
to test if the address 192.168.20.20 is accessible.

If you are unable to establish lossless communication, check the wiring, the network and the corresponding settings.



```
C:\WINNT\system32\cmd.exe
Microsoft Windows 2000 [Version 5.00.2195]
(C) Copyright 1985-2000 Microsoft Corp.

C:\>ping 192.168.20.20

Ping wird ausgeführt für 192.168.20.20 mit 32 Bytes Daten:

Antwort von 192.168.20.20: Bytes=32 Zeit<10ms TTL=128
Antwort von 192.168.20.20: Bytes=32 Zeit<10ms TTL=128
Antwort von 192.168.20.20: Bytes=32 Zeit<10ms TTL=128
Antwort von 192.168.20.20: Bytes=32 Zeit<10ms TTL=128

Ping-Statistik für 192.168.20.20:
    Pakete: Gesendet = 4, Empfangen = 4, Verloren = 0 (0% Verlust),
    Ca. Zeitangaben in Millisek.:
        Minimum = 0ms, Maximum = 0ms, Mittelwert = 0ms

C:\>_
```